



European Technical Assessment

ETA-26/0367 of 30/06/2026

English translation prepared by CSTB - Original version in French language

General Part

Technical Assessment Body issuing the European Technical Assessment:

Centre Scientifique et Technique du Bâtiment (CSTB)

**Trade name of the
construction product:**

Hilti angle connectors for MT-30 V2 and MT-40 V2 product ranges

**Product family to which the
construction product belongs:**

Products for installation systems for supporting technical building equipment

Manufacturer:

Hilti AG
Feldkircher Strasse 100
9494 Schaan
FÜRSTENTUM LIECHTENSTEIN

Manufacturing plants:

L 1005049, L 1124303, L 1027881, L 1087643

**This European Technical
Assessment contains:**

24 pages including 20 pages which form an integral part of this assessment

**This European Technical
Assessment is issued in
accordance with Article 95(4)
of Regulation (EU) 2024/3110,
on the basis of:**

European Assessment Document (EAD)
280016-00-0602 version June 2020

This version replaces:

Translations of this European Technical Assessment in other languages shall fully correspond to the original issued document and should be identified as such.

Communication of this European Technical Assessment, including transmission by electronic means, shall be in full (excepted the confidential Annex(es) referred to above). However, partial reproduction may be made, with the written consent of the issuing Technical Assessment Body. Any partial reproduction has to be identified as such.

Specific Parts

1 Technical description of the product

This European Technical Assessment covers HILTI angle connectors of MT System: MT-C-L1, MT-C-L1 OC, MT-C-Q1 FL, MT-C-Q1 FL OC, MT-C-L2, MT-C-L2 OC, MT-C-L2E FL, MT-C-L2E FL OC, MT-C-T/1, MT-C-T/1 OC, MT-C-LL1, MT-C-LL1 OC, MT-C-LL1 FL, MT-C-LL1 FL OC, MT-C-LL2, MT-C-LL2 OC, MT-C-LL2 FL, MT-C-LL2 FL OC, MT-C-T/1, MT-C-T/1 OC, MT-C-T/2, MT-C-T/2 OC, MT-C-T/2 FL, MT-C-T/2 FL OC, MT-C-GS OC, MT-CT-T, MT-CT-T OC, MT-C-T 3D/3 and MT-C-T 3D/3 OC.

MT-C-L1, MT-C-L1 OC, MT-C-Q1 FL and MT-C-Q1 FL OC angle connectors are made of zinc coated steel. The angle connectors have two leg angles of equal length, arranged at an angle of 90° between each other. One round opening is located centrally on each leg angle. MT-C-Q1 FL and MT-C-Q1 FL OC angle connectors are pre-assembled with channel connectors.

MT-C-L2, MT-C-L2 OC, MT-C-L2E FL and MT-C-L2E FL OC angle connectors are made of zinc coated steel. Two leg angles of different lengths are arranged at an angle of 90° between each other. One round opening is located centrally on shorter leg angle. Two round openings are located on the longer leg angle. MT-C-L2E FL and MT-C-L2E FL OC angle connectors are pre-assembled with channel connectors.

MT-C-T/1 and MT-C-T/1 OC angle connectors are made of zinc coated steel. Each connector arranges four parts to a combined U-shaped / L-shaped with one opening on each part.

MT-C-T/2, MT-C-T/2 OC, MT-C-T/2 FL and MT-C-T/2 FL OC angle connectors are made of zinc coated steel. Each connector combines two L-shaped 90° angles with two openings on each leg angle. The two L-shaped angles are arranged together in a T-shape. MT-C-T/2 FL and MT-C-T/2 FL OC are pre-assembled with channel connectors.

MT-C-LL1 and MT-C-LL1 OC angle connectors are made of zinc coated steel. Each connector combines two L-shaped 90° angles with one opening on each leg angle. A triangular component is connecting two leg angles at 90°.

MT-C-LL2, MT-C-LL2 OC, MT-C-LL2 FL and MT-C-LL2 FL OC angle connectors are made of zinc coated steel. Each connector combines two L-shaped 90° angles with two openings on each leg angle. A triangular component is connecting two leg angles at 90°. MT-C-LL2 FL and MT-C-LL2 FL OC angle connectors are pre-assembled with channel connectors.

MT-C-T 3D/3 and MT-C-T 3D/3 OC angle connectors are made of zinc coated steel. Each connector is combining three L-shaped 90° angles with one opening on each leg angle. The three L-shaped angles are arranged together at 90° in U-shape.

MT-CT-T and MT-CT-T OC connectors are made of zinc coated flat steel in cross-shape with 4 round holes.

MT-C-GS OC angle connectors are made of zinc coated steel. Two leg angles are arranged at an angle of 90° between each other and with stiffening ribs. Various openings in different sizes and shapes are located on the leg angle.

The drawings, dimensions and materials of the Hilti angle connectors of MT System are given in Annex A.

2 Specification of the intended use(s) in accordance with the applicable European Assessment Document (hereinafter EAD)

The performances given in clause 3 are only valid if HILTI installation channels of MT System are in compliance with the specifications and conditions given in Annex B.

The provisions made in this European Technical Assessment are based on an assumed working life of the HILTI installation channels of MT System of 50 years when installed in the works (provided that the installation systems products are subject to appropriate installation). The indications given on the working life cannot be interpreted as a guarantee given by the producer or Technical Assessment Body but are to be regarded only as a means for choosing the right products in relation to the expected economically reasonable working life of the works.

In accordance with the European Assessment Document EAD 280016-00-0602, the product is intended to be used in:

- a) installations for the support of sprinkler kits,
- b) installations for the support of technical building equipment in general,
- c) installations for the support of pipes for the transportation of water not intended for human consumption,
- d) installations for the support of pipes for the transport of gas/fuel intended for the supply of building heating / cooling systems.

3 Performance of the product and references to the methods used for its assessment

3.1 Safety in case of fire (BWR 2)

No.	Essential characteristic	Performance
1	Reaction to fire	Class A1
2	Resistance under fire exposure	No performance determined

3.2 Safety and accessibility in use (BWR 4)

No.	Essential characteristic	Performance
3	Characteristic resistance	See Annex C

4 Assessment and verification of constancy of performance (AVCP) system applied, with reference to its legal base

In accordance with the European Assessment Document EAD 280016-00-0602, the following legal bases apply:

- In case of intended use a) specified in Section 2:
Commission Decision N° 1996/577/EC as amended by Commission Decision 2002/592/EC:
The system is 1
- In case of intended use b) specified in Section 2:
Commission Decision N° 1997/161/EC de la Commission:
The system is 2+
- In case of intended use c) specified in Section 2:
Commission Decision N° 1999/472/EC as amended by Commission Decision 2001/596/EC:
The system is 3
- In case of intended use d) specified in Section 2:
Commission Decision N° 1999/472/EC as amended by Commission Decision 2001/596/EC:
The system is 4

5 Technical details necessary for the implementation of the AVCP system, as provided for in the applicable EAD

The technical details necessary for the implementation of the system for the assessment and verification of constancy of performance are laid down in the control plan (confidential part of this European Technical Assessment) deposited at Centre Scientifique et Technique du Bâtiment.

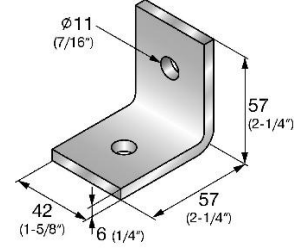
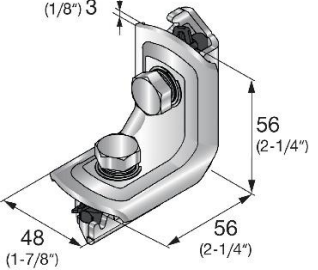
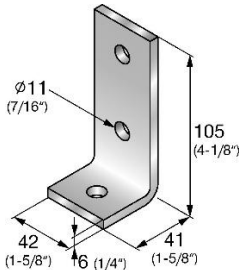
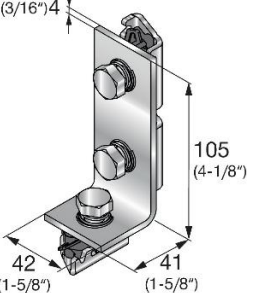
The manufacturer shall, on the basis of a contract, involve a notified body approved in the field of supporting systems for issuing the certificate of conformity CE based on the control plan.

The original French version is signed by

Loïc PAYET

Responsable de la division Structure, Maçonnerie et Partition

Table A1: Shape, dimensions and materials of MT-C-L1, MT-C-L1 OC, MT-C-Q1 FL, MT-C-Q1 FL OC, MT-C-L2, MT-C-L2 FL, MT-C-L2E FL and MT-C-L2E FL OC Hilti angle connectors

Item number	Designation	Materials and coatings	Illustration (Dimensions in mm)
2271514	MT-C-L1	Q355B acc. to GB/T 1591 galvanized	
2271516	MT-C-L1 OC	Q355B acc. to GB/T 1591 hot dip galvanized	
2399638	MT-C-Q1 FL	DD11 MOD acc. to EN1090 or equivalent as defined in HN555-4 galvanized	
2399673	MT-C-Q1 FL OC	DD11 MOD acc. to EN1090 or equivalent as defined in HN555-4 hot dip galvanized	
2271518	MT-C-L2	Q355B acc. to GB/T 1591 galvanized	
2271519	MT-C-L2 OC	Q355B acc. to GB/T 1591 hot dip galvanized	
2399666	MT-C-L2E FL	S235JR acc. to EN 10025-2 galvanized	
2399678	MT-C-L2E FL OC	S235JR acc. to EN 10025-2 hot dip galvanized	

Hilti angle connectors of MT-30 V2 and MT-40 V2 systems

Product description
Dimensions and materials

Annex A1

Table A2: Shape, dimensions and materials of MT-C-LL1, MT-C-LL1 OC, MT-C-LL1 FL, MT-C-LL1 FL OC, MT-CT-T, MT-CT-T OC, MT-C-T/1, MT-C-T/1 OC, MT-C-T 3D/3 and MT-C-T 3D/3 Hilti angle connectors

Item number	Designation	Materials and coatings	Illustration (Dimensions in mm)
2272047	MT-C-LL1	Q235B acc. to GB/T 700 galvanized	
2272049	MT-C-LL1 OC	Q235B acc. to GB/T 700 hot dip galvanized	
2399637	MT-C-LL1 FL	Q235B acc. to GB/T 700 galvanized	
2399672	MT-C-LL1 FL OC	Q235B acc. to GB/T 700 hot dip galvanized	
2322407	MT-CT-T	Q235B acc. to GB/T 700 galvanized	
2322411	MT-CT-T OC	Q235B acc. to GB/T 700 hot dip galvanized	
2272040	MT-C-T/1	Q355B acc. to GB/T 1591 galvanized	
2272042	MT-C-T/1 OC	Q355B acc. to GB/T 1591 hot dip galvanized	
2272060	MT-C-T 3D/3	Q355B acc. to GB/T 1591 galvanized	
2272061	MT-C-T 3D/3 OC	Q355B acc. to GB/T 1591 hot dip galvanized	

Hilti angle connectors of MT-30 V2 and MT-40 V2 systems

Product description
Dimensions and materials

Annex A2

Table A3: Shape, dimensions and materials of MT-C-LL2, MT-C-LL2 OC, MT-C-LL2 FL, MT-C-LL2 FL OC, MT-C-T/2, MT-C-T/2 OC, MT-C-T/2 FL and MT-C-T/2 OC Hilti angle connectors

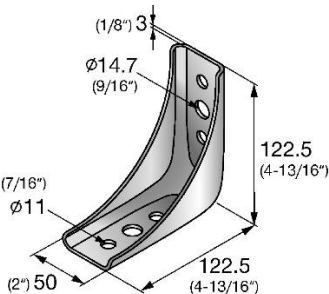
Item number	Designation	Materials and coatings	Illustration (Dimensions in mm)
2272051	MT-C-LL2	Q235B acc. to GB/T 700 galvanized	
2272053	MT-C-LL2 OC	Q235B acc. to GB/T 700 hot dip galvanized	
2399661	MT-C-LL2 FL	Q235B acc. to GB/T 700 galvanized	
2399676	MT-C-LL2 FL OC	Q235B acc. to GB/T 700 hot dip galvanized	
2272054	MT-C-T/2	Q235B acc. to GB/T 700 galvanized	
2272055	MT-C-T/2 OC	Q235B acc. to GB/T 700 hot dip galvanized	
2399668	MT-C-T/2 FL	Q235B acc. to GB/T 700 galvanized	
2399680	MT-C-T/2 FL OC	Q235B acc. to GB/T 700 hot dip galvanized	

Hilti angle connectors of MT-30 V2 and MT-40 V2 systems

Product description
Dimensions and materials

Annex A3

Table A4: Shape, dimensions and materials of MT-C-GS OC Hilti angle connectors

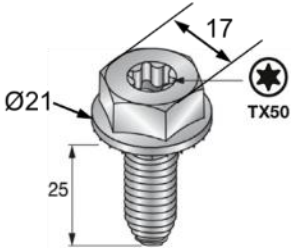
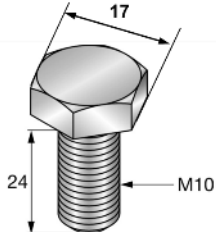
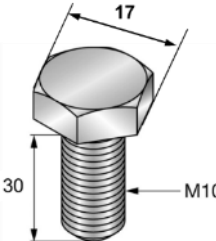
Item number	Designation	Materials and coatings	Illustration (Dimensions in mm)
2272064	MT-C-GS OC	Q355B acc. to GB/T 1591 hot dip galvanized	

Hilti angle connectors of MT-30 V2 and MT-40 V2 systems

Product description
Dimensions and materials

Annex A4

Table A5: Dimensions and materials of Hilti MT-TL M10, MT-TL M10 OC, MT-TFB OC, MT-TLB, MT-TLB OC, MT-TLB 30 and MT-TLB 30 OC channel connector

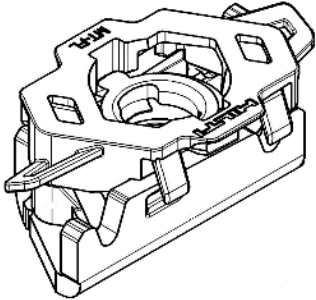
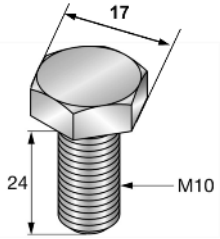
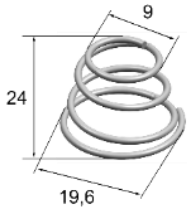
Item number	Designation	Materials and coatings	Illustration (Dimensions in mm)
2272080	MT-TL M10	<u>Steel part:</u> S460MC acc. to EN10149-2 or equivalent as defined in HN709 zinc coated <u>Plastic part:</u> Polyamid	
2272082	MT-TL M10 OC	<u>Steel part:</u> S460MC acc. to EN10149-2 or equivalent as defined in HN709 zinc coated with organic topcoat <u>Plastic part:</u> Polyamid	
2272084	MT-TFB OC	C10B21 acc. to SAE J403, Surface hardness min. 530 HV, Core hardness min. 32-39 HRC zinc coated with organic topcoat	
2273254	MT-TLB	strength class 8.8 acc. to EN ISO 898-1, zinc coated	
2273256	MT-TLB OC	strength class 8.8 acc. to EN ISO 898-1, zinc coated with organic topcoat	
2282190	MT-TLB 30	strength class 8.8 acc. to EN ISO 898-1, zinc coated	
2282191	MT-TLB 30 OC	strength class 8.8 acc. to EN ISO 898-1, zinc coated with organic topcoat	

Hilti angle connectors of MT-30 V2 and MT-40 V2 systems

Product description
Dimensions and materials

Annex A5

Table A6: Dimensions and material of Hilti MT-FL and MT-FL OC channel connectors

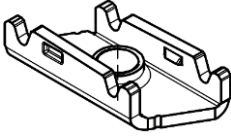
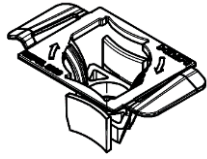
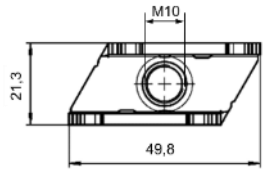
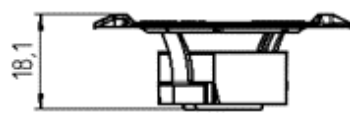
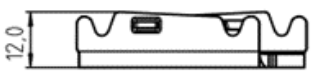
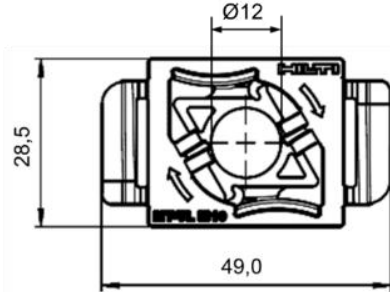
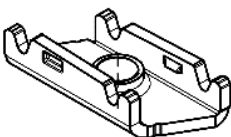
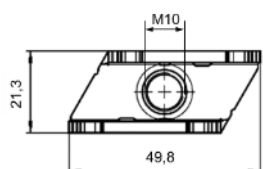
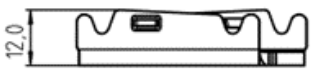
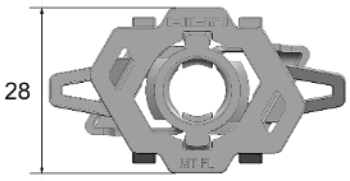
Item number	Designation	Materials and coatings	Illustration (Dimensions in mm)
2399683	MT-FL	<u>Steel part:</u> S460MC acc. to EN10149-2 or equivalent as defined in HN709 zinc coated <u>Plastic part:</u> Polypropylene	
2399682	MT-FL OC	<u>Steel part:</u> S460MC acc. to EN10149-2 or equivalent as defined in HN709 zinc coated with organic topcoat <u>Plastic part:</u> Polypropylene	
Additional components of MT-FL and MT-FL OC			
-	MT-FL	strength class 8.8 acc. to EN ISO 898-1, zinc coated	
-	MT-FL OC	strength class 8.8 acc. to EN ISO 898-1, zinc coated with organic topcoat	
-	MT-FL	1.4310 acc. to EN 10151	
-	MT-FL OC		

Hilti angle connectors of MT-30 V2 and MT-40 V2 systems

Product description
Dimensions and materials

Annex A6

Table A7: Dimensions of components of Hilti MT-TL M10, MT-TL M10 OC, MT-FL and MT-FL OC channel connectors

Item number	Designation	Illustration of nut (Dimensions in mm)	Illustration of plastic part (Dimensions in mm)
2272080 2272082	MT-TL M10 MT-TL M10 OC		
			
			
2399683 2399682	MT-FL MT-FL OC		
			
			

Hilti angle connectors of MT-30 V2 and MT-40 V2 systems

Product description
Dimensions and materials

Annex A7

Table A8: Dimensions and materials of Hilti MT-30 V2, MT-30 V2 S, MT-30 V2 OC, MT-30 V2 S OC, MT-30D V2, MT-30D V2 S, MT-30D V2 OC and MT-30D V2 S OC installation channels

Illustration (Dimensions in mm and inch)	Item number	Designation	Length [m]	Materials and coatings
	2465592	MT-30 V2 S	3	S280GD or equivalent as defined in HN704 + Z275-M-A-C acc. to EN 10346
	2465565	MT-30 V2	6	
	2465593	MT-30 V2 S OC	3	S280GD or equivalent as defined in HN704 + ZM310-A-C acc. to EN 10346
	2465566	MT-30 V2 OC	6	
Two profiles of MT-30 V2 channel are connected in the area of the holes in the back of the channels in a shape-fitting and force-fitting way as a kind of riveted connection.	2465558	MT-30D V2 S	3	S280GD or equivalent as defined in HN704 + Z275-M-A-C acc. to EN 10346
	2465790	MT-30D V2	6	
	2465582	MT-30D V2 S OC	3	S280GD or equivalent as defined in HN704 + ZM310-A-C acc. to EN 10346
	2465583	MT-30D V2 OC	6	

Hilti angle connectors of MT-30 V2 and MT-40 V2 systems

Product description
Dimensions and materials

Annex A8

Illustration (Dimensions in mm and inch)	Item number	Designation	Length [m]	Materials and coatings
	2465567	MT-40 V2 S	3	S280GD or equivalent as defined in HN704 + Z275-M-A-C acc. to EN 10346
	2465581	MT-40 V2	6	
	2465568	MT-40 V2 S OC	3	S280GD or equivalent as defined in HN704 + ZM310-A-C acc. to EN 10346
	2465569	MT-40 V2 OC	6	
	2465577	MT-40D V2 S	3	S280GD or equivalent as defined in HN704 + Z275-M-A-C acc. to EN 10346
	2465791	MT-40D V2	6	
	2465792	MT-40D V2 S OC	3	S280GD or equivalent as defined in HN704 + ZM310-A-C acc. to EN 10346
	2465578	MT-40D V2 OC	6	

Two profiles of MT-40 V2 channel are connected in the area of the holes in the back of the channels in a shape-fitting and force-fitting way as a kind of riveted connection.

Product description

Dimensions and materials

Annex A9

Specifications of intended use

- HILTI angle connectors of MT System are used to transfer building services components loads such as ducts and equipment for water, heating, cooling, ventilation, electrical and other systems.
- HILTI MT angle connectors are suitable for undertaking this load-bearing function under conditions described in Section 2 of this European Technical Assessment.
- The resistance of HILTI angle connectors set down in Annex C applies for static actions in the direction of the main axes X, Y, Z in connection with HILTI installation channels described in Annex A8 and A9 and in combination with HILTI channel connectors Annex A1 to A4.
- The installation torque of channel connectors are shown in Table B1

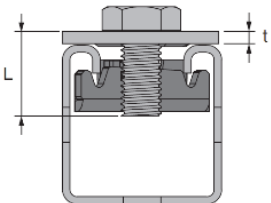
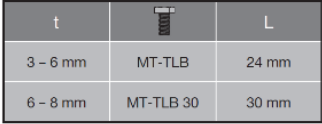
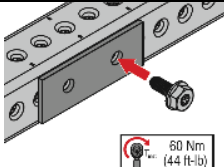
HILTI channel connectors	Bolts	Torque	Illustration
MT-TL M10	MT-TLB MT-TLB 30	30 Nm	
MT-FL	n.a.		
MT-TL M10 OC	MT-TLB OC MT-TLB 30 OC	40 Nm	
MT-FL OC	n.a.		
MT-TFB OC	n.a.	60 Nm	

Table B1: Installation torque of MT channel connectors

- The required torque may be applied with electrical or non-electrical devices.
- The MT installation open profile channels and closed profiles (girders) can be cut along the entire length following the manufacturers' instructions without compromising the declared performances.
- For closed MT channels (girders) the distance between the cut of the girder and start of the dome shaped hole must be at a minimum distance of 10 mm.
- Prior to installation, it must be ensured that the supported component, the anchoring to the base material and the base material itself are suitable to withstand the resistance values of the installation system. The angle connectors must be installed by appropriately qualified personnel and under the supervision of the site manager. The general installation instructions of the manufacturer apply.

Hilti angle connectors of MT-30 V2 and MT-40 V2 systems

Intended use
Specifications

Annex B1

Table C1: Characteristic resistance of the angle connectors MT-C-L1 and MT-C-L1 OC in combination with installation channels and channel connectors

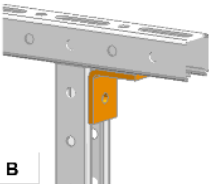
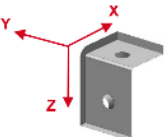
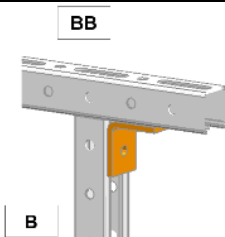
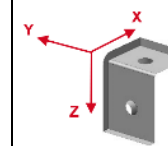
System specific configuration	Static action directions	HILTI angle connector (all coatings)	Installation channel (all lengths and coatings)	Number and type of HILTI channel connectors (all coatings)		
BB  B		MT-C-L1	MT-30 V2 MT-40 V2 MT-30D V2 MT-40D V2	2 x MT-TL M10 or 2 x MT FL		
Characteristic resistances in kN						
Installation channel	+ F _{x,Rk}	- F _{x,Rk}	+ F _{y,Rk}	- F _{y,Rk}	+ F _{z,Rk}	- F _{z,Rk}
MT-30 V2	5,65	5,89	NPA	NPA	5,65	5,89
MT-30D V2	4,95	5,82	NPA	NPA	4,95	5,82
MT-40 V2	9,34	8,08	NPA	NPA	9,34	8,08
MT-40D V2	9,34	8,08	NPA	NPA	9,34	8,08

Table C2: Characteristic resistance of the angle connectors MT-C-Q1 FL and MT-C-Q1 FL OC in combination with installation channels and channel connectors

System specific configuration	Static action directions	HILTI angle connector (all coatings)	Installation channel (all lengths and coatings)	Number and type of HILTI channel connectors (all coatings)		
BB  B		MT-C-Q1 FL	MT-30 V2, MT-30D V2	Item pre-assembled with MT-FL		
Characteristic resistances in kN						
Installation channel	+ F _{x,Rk}	- F _{x,Rk}	+ F _{y,Rk}	- F _{y,Rk}	+ F _{z,Rk}	- F _{z,Rk}
MT-30 V2	5,24	4,72	NPA	NPA	5,24	4,72
MT-30D V2	5,24	4,72	NPA	NPA	5,24	4,72
MT-40 V2	7,71	6,17	NPA	NPA	7,71	6,17
MT-40D V2	7,71	6,17	NPA	NPA	7,71	6,17

Hilti angle connectors of MT-30 V2 and MT-40 V2 systems

Characteristic performance

Annex C1

Table C3: Characteristic resistance of the angle connectors MT-C-L2 and MT-C-L2 OC in combination with installation channels and channel connectors

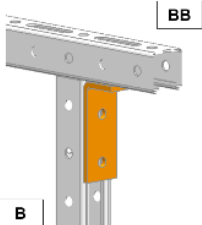
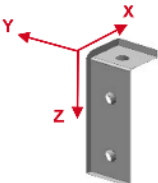
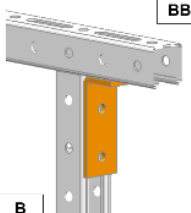
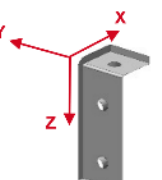
System specific configuration	Static action directions	HILTI angle connector (all coatings)	Installation channel (all lengths and coatings)	Number and type of HILTI channel connectors (all coatings)		
		MT-C-L2	MT-40 V2 MT-40D V2 MT-30 V2 MT-30D V2	3 x MT-TL M10 or 3 x MT-FL		
Characteristic resistances in kN						
Installation channel	+ F _{x,Rk}	- F _{x,Rk}	+ F _{y,Rk}	- F _{y,Rk}	+ F _{z,Rk}	- F _{z,Rk}
MT-30 V2	NPA	NPA	NPA	NPA	5,84	5,26
MT-30D V2	NPA	NPA	NPA	NPA	10,92	9,83
MT-40 V2	NPA	NPA	NPA	NPA	10,06	9,84
MT-40D V2	NPA	NPA	NPA	NPA	10,06	9,84

Table C4: Characteristic resistance of the angle connectors MT-C-L2E FL and MT-C-L2E FL OC in combination with installation channels and channel connectors

System specific configuration	Static action directions	HILTI angle connector (all coatings)	Installation channel (all lengths and coatings)	Number and type of HILTI channel connectors (all coatings)		
		MT-C-L2E FL	MT-30 V2 MT-30D V2 MT-40 V2 MT-40D V2	Item pre-assembled with MT-FL		
Characteristic resistances in kN						
Installation channel	+ F _{x,Rk}	- F _{x,Rk}	+ F _{y,Rk}	- F _{y,Rk}	+ F _{z,Rk}	- F _{z,Rk}
MT-30 V2	NPA	NPA	NPA	NPA	4,96	4,46
MT-30D V2	NPA	NPA	NPA	NPA	6,87	6,18
MT-40 V2	NPA	NPA	NPA	NPA	8,10	6,48
MT-40D V2	NPA	NPA	NPA	NPA	8,10	6,48

Hilti angle connectors of MT-30 V2 and MT-40 V2 systems

Characteristic performance

Annex C2

Table C5: Characteristic resistance of the angle connectors MT-C-LL1, MT-C-LL1 OC, MT-C-LL1 FL and MT-C-LL1 FL OC in combination with installation channels and channel connectors

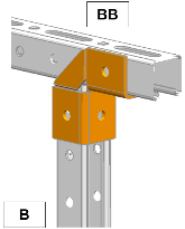
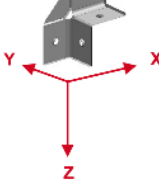
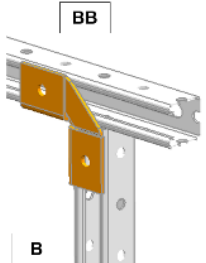
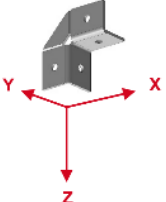
System specific configuration	Static action directions	HILTI angle connector (all coatings)	Installation channel (all lengths and coatings)	Number and type of HILTI channel connectors (all coatings)		
		MT-C-LL1 or MT-C-LL1 FL*)	MT-30 V2 MT-40 V2 MT-30D V2 MT-40D V2	2 x MT-TL M10 or 2 x MT-FL *)Item pre-assembled with MT-FL		
Characteristic resistances in kN						
Installation channel	+ F _{x,Rk}	- F _{x,Rk}	+ F _{y,Rk}	- F _{y,Rk}	+ F _{z,Rk}	- F _{z,Rk}
MT-30 V2	4,90	4,54	NPA	NPA	4,90	4,54
MT-30D V2	6,87	6,19	NPA	NPA	6,87	6,19
MT-40 V2	8,60	7,13	NPA	NPA	8,60	7,13
MT-40D V2	8,60	7,13	NPA	NPA	8,60	7,13

Table C6: Characteristic resistance of the angle connectors MT-C-LL1 and MT-C-LL1 OC in combination with installation channels and channel connectors

System specific configuration	Static action directions	HILTI angle connector (all coatings)	Installation channel (all lengths and coatings)	Number and type of HILTI channel connectors (all coatings)		
		MT-C-LL1	MT-40 V2 MT-40D V2	2 x MT-TL M10 or 2 x MT-FL		
Characteristic resistances in kN						
Installation channel	+ F _{x,Rk}	- F _{x,Rk}	+ F _{y,Rk}	- F _{y,Rk}	+ F _{z,Rk}	- F _{z,Rk}
MT-40 V2	4,77	3,78	NPA	NPA	4,77	3,78
MT-40D V2	4,77	3,78	NPA	NPA	4,77	3,78

Hilti angle connectors of MT-30 V2 and MT-40 V2 systems

Characteristic performance

Annex C3

Table C7: Characteristic resistance of the angle connectors MT-C-LL2, MT-C-LL2 OC, MT-C-LL2 FL and MT-C-LL2 FL OC in combination with installation channels and channel connectors

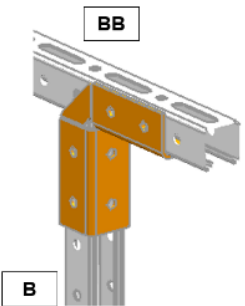
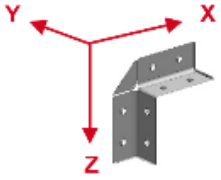
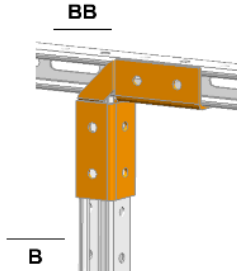
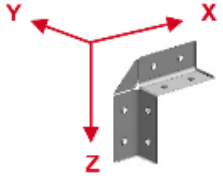
System specific configuration	Static action directions	HILTI angle connector (all coatings)	Installation channel (all lengths and coatings)	Number and type of HILTI channel connectors (all coatings)		
		MT-C-LL2 or MT-C-LL2 FL ^{*)}	MT-40 V2 MT-40D V2 MT-30 V2 MT-30D V2	4 x MT-TL M10 or 4 x MT-FL ^{*)} Item pre-assembled with MT-FL		
Characteristic resistances in kN						
Installation channel	+ F _{x,Rk}	- F _{x,Rk}	+ F _{y,Rk}	- F _{y,Rk}	+ F _{z,Rk}	- F _{z,Rk}
MT-30 V2	4,06	3,79	NPA	NPA	4,06	3,79
MT-30D V2	7,01	6,31	NPA	NPA	7,01	6,31
MT-40 V2	8,22	6,71	NPA	NPA	8,22	6,71
MT-40D V2	10,26	6,66	NPA	NPA	10,26	6,66

Table C8: Characteristic resistance of the angle connectors MT-C-LL2 and MT-C-LL2 OC in combination with installation channels and channel connectors

System specific configuration	Static action directions	HILTI angle connector (all coatings)	Installation channel (all lengths and coatings)	Number and type of HILTI channel connectors (all coatings)		
		MT-C-LL2	MT-40 V2, MT-40D V2	4 x MT-TL M10 or 4 x MT-FL		
Characteristic resistances in kN						
Installation channel	+ F _{x,Rk}	- F _{x,Rk}	+ F _{y,Rk}	- F _{y,Rk}	+ F _{z,Rk}	- F _{z,Rk}
MT-40 V2	5,43	6,03	NPA	NPA	5,43	6,03
MT-40D V2	5,43	6,03	NPA	NPA	5,43	6,03

Hilti angle connectors of MT-30 V2 and MT-40 V2 systems

Characteristic performance

Annex C4

Table C9: Characteristic resistance of the angle connectors MT-C-T/2, MT-C-T/2 OC, MT-C-T/2 FL, MT-C-T/2 FL OC in combination with channels and channel connectors

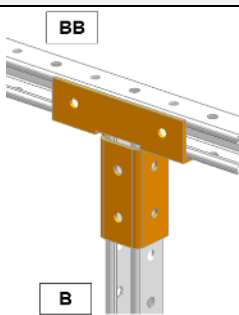
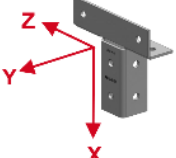
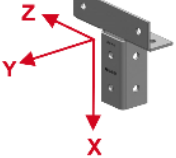
System specific configuration	Static action directions	HILTI angle connector (all coatings)	Installation channel (all lengths and coatings)	Number and type of HILTI channel connectors (all coatings)		
		MT-C-T/2 or MT-C-T/2 FL ^{*)}	MT-30D V2	4 x MT-TL M10 or 4 x MT-FL ^{*)} Item pre-assembled with MT-FL		
Characteristic resistances in kN						
Installation channel	+ F _{x,Rk}	- F _{x,Rk}	+ F _{y,Rk}	- F _{y,Rk}	+ F _{z,Rk}	- F _{z,Rk}
MT-30D V2	NPA	NPA	NPA	NPA	7,47	7,47

Table C10: Characteristic resistance of the angle connectors MT-C-T/2 and MT-C-T/2 OC in combination with channels and channel connectors

System specific configuration	Static action directions	HILTI angle connector (all coatings)	Installation channel (all lengths and coatings)		Number and type of HILTI channel connectors (all coatings)	
		MT-C-T/2	MT-30 V2		4 x MT-TL M10 or 4 x MT-FL	
Characteristic resistances in kN						
Installation channel	+ F _{x,Rk}	- F _{x,Rk}	+ F _{y,Rk}	- F _{y,Rk}	+ F _{z,Rk}	- F _{z,Rk}
MT-30 V2	NPA	NPA	NPA	NPA	3,48	4,09
MT-40 V2	NPA	NPA	NPA	NPA	8,23	6,58

Hilti angle connectors of MT-30 V2 and MT-40 V2 systems

Characteristic performance

Annex C5

Table C11: Characteristic resistance of the angle connectors MT-C-GS OC in combination with installation channels and channel connectors

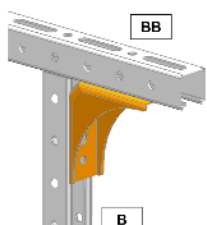
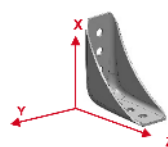
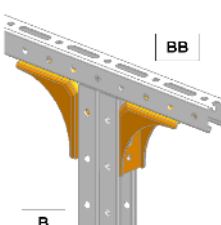
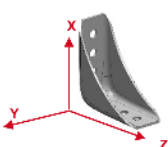
System configuration	Static action directions	HILTI angle connector (all coatings)	Installation channel (all lengths and coatings)	Number and type of HILTI channel connectors (all coatings)		
		MT-C-GS OC	MT-30 V2 MT-30D V2 MT-40 V2 MT-40D V2	4 x MT-TL M10 or 4 x MT-FL		
Characteristic resistances in kN						
Installation channel	+ $F_{x,Rk}$	- $F_{x,Rk}$	+ $F_{y,Rk}$	- $F_{y,Rk}$	+ $F_{z,Rk}$	- $F_{z,Rk}$
MT-30 V2	4,25	3,83	NPA	NPA	4,25	3,83
MT-30D V2	7,16	6,44	NPA	NPA	7,16	6,44
MT-40 V2	10,29	8,23	NPA	NPA	10,29	8,23
MT-40D V2	14,93	8,23	NPA	NPA	14,93	8,23

Table C12: Characteristic resistance of the angle connectors MT-C-GS OC in combination with installation channels and channel connectors

System specific configuration	Static action directions	HILTI angle connector (all coatings)	Installation channel (all lengths and coatings)	Number and type of HILTI channel connectors (all coatings)		
		2 x MT-C-GS OC	Channel B: MT-30D V2 MT-40D V2 Channel BB : MT-30 V2 MT-30D V2 MT-40 V2 MT-40D V2	8 x MT-TL M10 OC or 8 x MT-FL OC		
Characteristic resistances in kN						
Installation channel	+ F _{x,Rk}	- F _{x,Rk}	+ F _{y,Rk}	- F _{y,Rk}	+ F _{z,Rk}	- F _{z,Rk}
MT-30 V2	4,25	3,83	NPA	NPA	4,25	3,83
MT-30D V2	7,16	6,44	NPA	NPA	7,16	6,44
MT-40 V2	10,29	8,23	NPA	NPA	10,29	8,23
MT-40D V2	14,93	8,23	NPA	NPA	14,93	8,23

Hilti angle connectors of MT-30 V2 and MT-40 V2 systems

Characteristic performance

Annex C6

Table C13: Characteristic resistance of the angle connector MT-C-GS OC in combination with installation channels and channel connectors

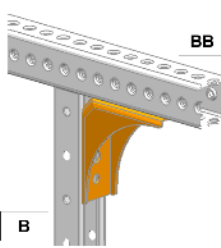
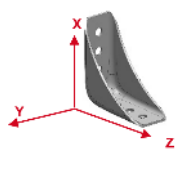
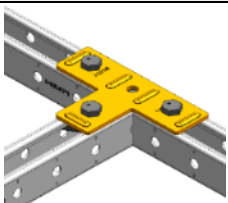
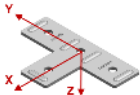
System specific configuration	Static action directions	HILTI angle connector (all coatings)	Installation channel (all lengths and coatings)		Number and type of HILTI channel connectors (all coatings)	
		MT-C-GS OC	Channel B: MT-30 V2, MT-30D V2 MT-40 V2, MT-40D V2 Channel BB: MT-70 OC MT-80 OC MT-90 OC MT-100 OC		2 x MT-TL M10 OC or 2 x MT-FL OC and 2 x MT-TFB OC	
Characteristic resistances in kN						
Installation channel	+ F _{x,Rk}	- F _{x,Rk}	+ F _{y,Rk}	- F _{y,Rk}	+ F _{z,Rk}	- F _{z,Rk}
MT-30 V2	4,25	3,83	NPA	NPA	4,25	3,83
MT-30D V2	7,16	6,44	NPA	NPA	7,16	6,44
MT-40 V2	10,29	8,23	NPA	NPA	10,29	8,23
MT-40D V2	14,93	8,23	NPA	NPA	14,93	8,23

Table C14: Characteristic resistance of the angle connectors MT-CT-T and MT-CT-T OC in combination with channels and channel connectors

System specific configuration	Static action directions	HILTI angle connector (all coatings)	Installation channel (all lengths and coatings)		Number and type of HILTI channel connectors (all coatings)	
		MT-CT-T	MT-40 V2		3 x MT-TL M10 or 3 x MT-FL	
Characteristic resistances in kN						
Installation channel	+ F _{x,Rk}	- F _{x,Rk}	+ F _{y,Rk}	- F _{y,Rk}	+ F _{z,Rk}	- F _{z,Rk}
MT-40 V2	NPA	NPA	2,80	2,80	NPA	NPA

Hilti angle connectors of MT-30 V2 and MT-40 V2 systems

Characteristic performance

Annex C7

Table C15: Characteristic resistance of the angle connectors MT-CT-T and MT-CT-T OC in combination with channels and channel connectors

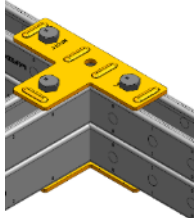
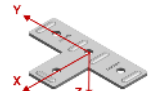
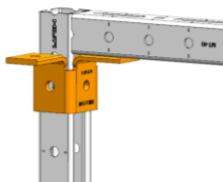
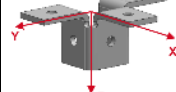
System specific configuration	Static action directions	HILTI angle connector (all coatings)	Installation channel (all lengths and coatings)		Number and type of HILTI channel connectors (all coatings)	
		2 x MT-CT-T	MT-40D V2		6 x MT-TL M10 or 6 x MT-FL	
Characteristic resistances in kN						
Installation channel	+ F _{x,Rk}	- F _{x,Rk}	+ F _{y,Rk}	- F _{y,Rk}	+ F _{z,Rk}	- F _{z,Rk}
MT-40D V2	NPA	NPA	9,77	9,77	NPA	NPA

Table C16: Characteristic resistance of the angle connectors MT-C-T 3D/3 and MT-C-T 3D/3 OC in combination with channels and channel connectors

System specific configuration	Static action directions	HILTI angle connector (all coatings)	Installation channel (all lengths and coatings)		Number and type of HILTI channel connectors (all coatings)	
		MT-C-T 3D/3	MT-30D V2 MT-40 V2 MT-40D V2		2 x MT-TL M10 or 2 x MT-FL	
Characteristic resistances in kN						
Installation channel	+ F _{x,Rk}	- F _{x,Rk}	+ F _{y,Rk}	- F _{y,Rk}	+ F _{z,Rk}	- F _{z,Rk}
MT-30D V2	NPA	NPA	NPA	NPA	7,68	6,91
MT-40 V2	NPA	NPA	NPA	NPA	9,89	7,91
MT-40D V2	NPA	NPA	NPA	NPA	9,89	7,91

Hilti angle connectors of MT-30 V2 and MT-40 V2 systems

Characteristic performance

Annex C8

Table C17: Characteristic resistance of the angle connectors MT-C-T 3D/3 and MT-C-T 3D/3 OC in combination with channels and channel connectors

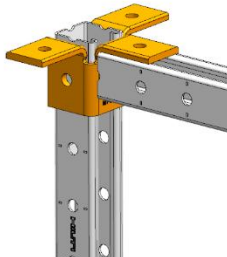
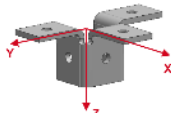
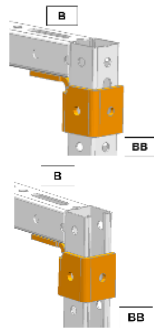
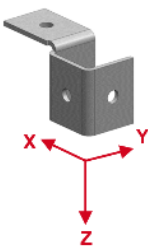
System specific configuration	Static action directions	HILTI angle connector (all coatings)	Installation channel (all lengths and coatings)	Number and type of HILTI channel connectors (all coatings)		
		MT-C-T 3D/3	MT-40 V2 MT-40D V2	2 x MT-TL M10 or 2 x MT-FL		
Characteristic resistances in kN						
Installation channel	+ F _{x,Rk}	- F _{x,Rk}	+ F _{y,Rk}	- F _{y,Rk}	+ F _{z,Rk}	- F _{z,Rk}
MT-40 V2	NPA	NPA	NPA	NPA	6,56	8,20
MT-40D V2	NPA	NPA	NPA	NPA	6,56	8,20

Table C18: Characteristic resistance of the angle connectors MT-C-T/1 and MT-C-T/1 OC in combination with installation channels and channel connectors

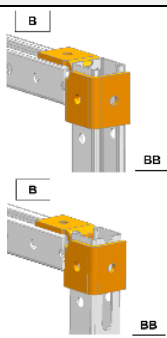
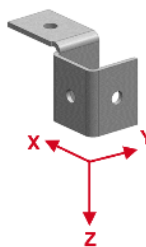
System specific configuration	Static action directions	HILTI angle connector (all coatings)	Installation channel (all lengths and coatings)	Number and type of HILTI channel connectors (all coatings)		
		MT-C-T/1	Channel B: MT-40 V2 MT-40D V2 MT-30D V2 Channel BB: MT-40 V2 MT-30D V2	2 x MT-TL M10 or 2 x MT-FL		
Characteristic resistances in kN						
Installation channel	+ F _{x,Rk}	- F _{x,Rk}	+ F _{y,Rk}	- F _{y,Rk}	+ F _{z,Rk}	- F _{z,Rk}
MT-30D V2	NPA	NPA	NPA	NPA	7,68	6,91
MT-40 V2	NPA	NPA	NPA	NPA	9,89	7,91
MT-40D V2	NPA	NPA	NPA	NPA	9,89	7,91

Hilti angle connectors of MT-30 V2 and MT-40 V2 systems

Characteristic performance

Annex C9

Table C19: Characteristic resistance of the angle connectors MT-C-T/1 and MT-C-T/1 OC in combination with installation channels and channel connectors

System specific configuration	Static action directions	HILTI angle connector (all coatings)	Installation channel (all lengths and coatings)	Number and type of HILTI channel connectors (all coatings)		
		MT-C-T/1	Channel B: MT-40 V2 MT-40D V2 MT-30D V2 Channel BB: MT-40 V2 MT-30D V2	2 x MT-TL M10 or 2 x MT-FL		
Characteristic resistances in kN						
Installation channel	+ F _{x,Rk}	- F _{x,Rk}	+ F _{y,Rk}	- F _{y,Rk}	+ F _{z,Rk}	- F _{z,Rk}
MT-40 V2	NPA	NPA	NPA	NPA	6,56	8,20
MT-40D V2	NPA	NPA	NPA	NPA	6,56	8,20

Designation

$F_{x,R}$ Resistance in direction of local x-Axis
 $F_{y,R}$ Resistance in direction of local y-Axis
 $F_{z,R}$ Resistance in direction of local z-Axis

$M_{x,R}$ Resistance around local x-Axis
 $M_{y,R}$ Resistance around local y-Axis
 $M_{z,R}$ Resistance around local z-Axis

All characteristic resistances for ambient temperatures do not consider deflections,
 Partial safety factor for design resistance is $\gamma_M = F_{Rk} / F_{Rd}$ or $\gamma_M = M_{Rk} / M_{Rd}$,
 For design resistances the manufacturer's specifications and national regulations must be observed,

Hilti angle connectors of MT-30 V2 and MT-40 V2 systems

Characteristic performance

Annex C10